Kishorekumar Avudai Nayagam

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Summary

Software Engineer with 3+ years of experience specializing in scalable full-stack and AI systems, building RAG pipelines, agentic workflows, MCP-powered tool servers, and cloud-native microservices. Delivered measurable production impact by reducing support tickets by **80%** and release cycles by **40%**. Seeking Software Engineer or AI Engineer roles focused on AI infrastructure and distributed systems.

Education

University at Buffalo, State University of New York — Aug 2024 – Dec 2025
Master of Science in Computer Science — *Buffalo, NY*

Technical Skills

Languages: Python, JavaScript, TypeScript, Java, SQL, Scala, Bash
Frontend & Backend: React.js, Next.js, Node.js, FastAPI, Spring Boot, REST APIs, Microservices
AI & Agentic Systems: LangChain, LangGraph, OpenAI API, Claude API, RAG, FAISS, Pinecone, MCP
Cloud & DevOps: AWS (EC2, S3, Lambda, DynamoDB), GCP, Docker, Kubernetes, GitHub Actions, CI/CD
Databases: PostgreSQL, MongoDB, Redis, MySQL
Tools: Jest, Pytest, Postman, Kafka, Spark, Tableau, JIRA, GitHub Copilot, Claude Code

Work Experience

Research Foundation at SUNY — Feb 2026 – Present
Research Software Engineer | *React, FastAPI, PostgreSQL, LangGraph, Gemini API, MCP, AWS*

- Designed and built a full-stack GIS research platform using React, FastAPI, and PostgreSQL to model geospatial entities and temporal metadata, cataloging **500+ buildings** and **15+ years** of Google Street View imagery.
- Engineered an agentic AI analysis pipeline with LangGraph, orchestrating tools for coordinate sampling, Street View retrieval, Gemini-based facade scoring, and result persistence via MCP tool servers with a RAG layer for context-aware detection.
- Implemented a road-aligned coordinate sampling algorithm using the Google Directions API with FastAPI async pipelines secured via JWT and RBAC, reducing analysis latency by **75%** and manual assessment time by **80%**.
- Optimized serverless deployment on Vercel using Python asyncio and pre-warming strategies, eliminating cold-start latency and reducing end-to-end response time by **75%** at scale.

Cognizant Technology Solutions — Jan 2022 – May 2024
Software Engineer | *React.js, Node.js, PostgreSQL, MongoDB, Jest, Docker, GitHub Actions*

- Modernized a legacy monitoring platform from jQuery + JSP to a modular React.js architecture, enabling real-time dashboards for **10,000+** daily enterprise users and reducing customer support tickets by **80%**.
- Designed RESTful APIs using Node.js with PostgreSQL and MongoDB, implementing schema validation, indexing, and query optimization, reducing retrieval latency by **30%** and cloud infrastructure costs by **15%**.
- Improved reliability through Jest-based test suites and CI/CD pipelines using GitHub Actions and Docker, preventing regressions across **10+** production releases and shortening release cycles by **40%**.
- Engineered backend services integrating third-party systems with idempotent processing, schema validation, and retry logic to eliminate sync lag and schema-related failures.

Projects

LLM Inference Serving System | Python, FastAPI, PyTorch, asyncio, Prometheus, Docker | GitHub

- Built a production-grade LLM inference server from scratch implementing an Orca-style continuous batching engine — requests are admitted, prefilled, and decoded each iteration tick rather than waiting for full batch completion, eliminating head-of-line blocking.
- Designed a block-based KV cache manager with LRU eviction modeled after OS virtual memory paging, implementing the PagedAttention block table abstraction to eliminate memory fragmentation across concurrent requests.
- Instrumented with Prometheus metrics (queue depth, batch efficiency, time-to-first-token, cache utilization) with pre-wired Grafana dashboard; **229 unit and integration tests** across all 10 system layers.

Codebase Intelligence Platform | React, FastAPI, Neo4j, PostgreSQL, Redis, Docker, AWS | GitHub

- Built a full-stack developer tool with a React frontend and FastAPI backend, modeling GitHub repository structure as a Neo4j property graph to enable dependency mapping, security scanning, and semantic search across **10,000+** source files with sub-second query response.
- Designed a REST API layer with JWT authentication and Redis caching, containerized with Docker and deployed on AWS EC2, enabling scalable low-latency access; augmented query results with an LLM-powered RAG layer for context-aware code explanation.

Agentic Support Knowledge Platform | React, FastAPI, PostgreSQL, FAISS, Docker, AWS | GitHub

- Built a full-stack document search platform with a React UI, FastAPI backend, and PostgreSQL metadata store, serving **200+** concurrent users on AWS EC2 with async request handling and Prometheus-monitored pipelines.
- Improved search relevance from **67% to 92%** by implementing a hybrid retrieval layer combining FAISS vector search with cross-encoder re-ranking, and reduced incorrect results by **35%** through iterative LangGraph validation agents.